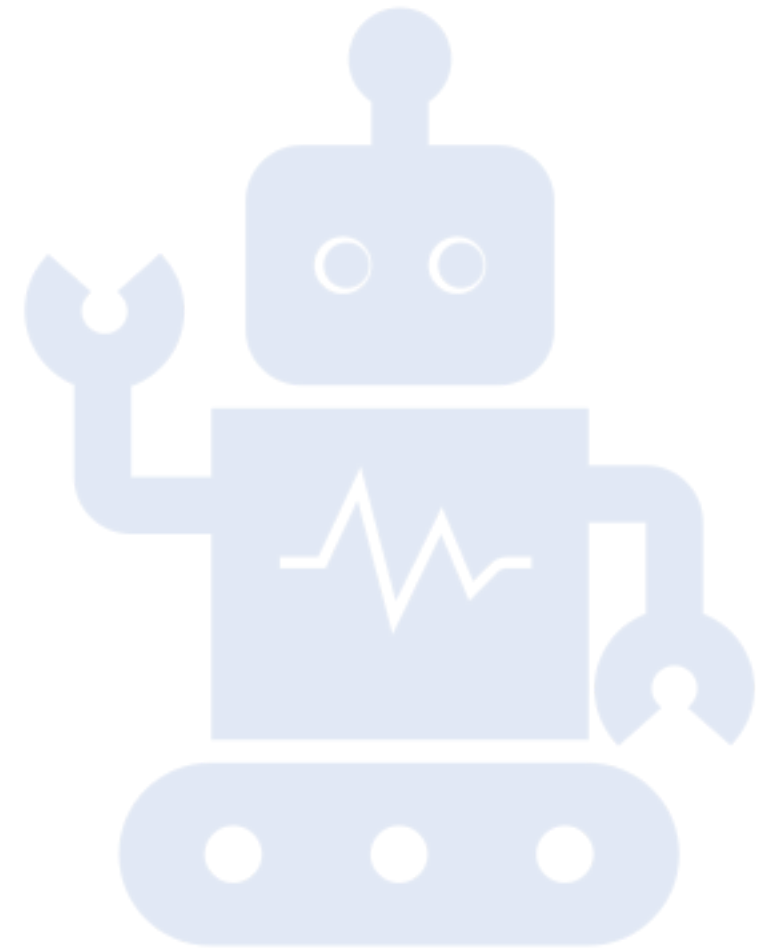




**JASHORE UNIVERSITY OF
SCIENCE AND TECHNOLOGY**

CSE 3201

Artificial Intelligence and Machine

Learning Lecture 1 - Introduction

Dr. A F M Shahab Uddin
Assistant Professor

Department of Computer Science and Engineering
Jashore University of Science and Technology

What can we achieve with State-of-the-art AI / ML?

- **Teach a robot to interact with
complex objects**



3

What can we achieve with State-of-the-art AI / ML?

- Teach a robot to interact with complex objects



- Teach a computer to play complex games



4

What can we achieve with State-of-the-art AI / ML?

- Teach a robot to interact with complex objects

- Teach a computer to play complex games
- **Teach a computer to describe images**

Image from: <https://github.com/danieljl/keras-image-captioning> 5

What can we achieve with State-of-the-art AI / ML?

- Teach a robot to interact with



complex objects

- Teach a computer to play complex games
- Teach a computer to describe images
- **Robots that adapt**

Autonomous vehicles



Aftenposten 2022-01-09, Arkivfoto: Siri Øverland EriksenNew York Times, Nov 21, 2023

Speech text



und Polarisation des Lichtes in den sogenannten kolloidalen Metallösungen. Zieht man die Folgerungen aus der elektromagnetischen Lichttheorie auf das Verhalten der trüben Medien, so kommt man zu verschiedenen Resultaten, je nachdem die trübenden Teilchen Isolatoren oder Leiter der Elektrizität sind. Die bezüglichen Rechnungen sind durchgeführt worden von Lord Rayleigh¹⁾ für Isolatoren und von J. J. Thomson²⁾ für Leiter der Elektrizität. Dabei machen beide die Annahme, daß die kleinen Teilchen Kugeln mit gegen die Lichtwellenlänge kleinem Durchmesser sind. Beide Autoren behandeln das Problem der Zerstreuung des Lichtes durch eine solche kleine Kugel, wenn diese von einer Welle natürlichen Lichtes getroffen wird. Während nun die Rechnung ergab, daß das von einer isolierenden Kugel in einer Ebene senkrecht zum einfallenden Strahl zerstreute Licht vollkommen linear polarisiert ist, und zwar in der durch die betrachtete Zerstreuungsrichtung



8

Machine translation

9

Introduction to Artificial Intelligence
Problem Solving and Search
Knowledge Representation and Reasoning



Introduction to Machine Learning
Supervised Learning
Unsupervised Learning

Neural Networks and Deep Learning (Intro)

Reinforcement Learning (Conceptual Overview)

AI and ML Applications

[Source: Wikipedia](#)

The first use of the term "Artificial Intelligence"

- **Dartmouth summer research project on artificial intelligence: US, 1956**

- Organized by **John McCarthy, Marvin Minsky, Nathaniel Rochester, Claude Shannon**

- "The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of

intelligence can in principle be so precisely described that a machine can be made to simulate it."

- "An attempt will be made to find how to make machines use language, form abstractions and concepts, solve

kinds of problems now reserved for humans, and improve themselves."

- "We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer. (2-month, 10 man)"

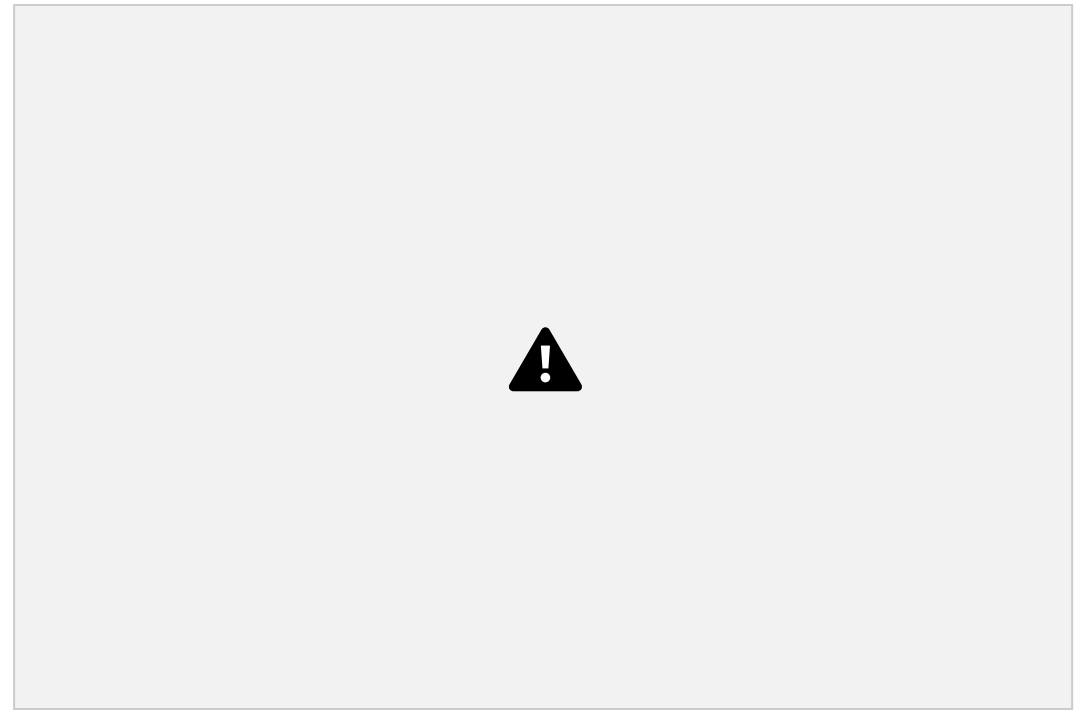


Source:

What is Intelligence?

(in humans:)

- No consensus definition,
- but • "Mental abilities"
 - Aptitude – not acquired knowledge
 - More than one dimension



- [Source](#) • Are **animals** intelligent?
- Various lists of what to include
- Emphasis on differences
 - Can **machines** be intelligent?
 - Can a **chatbot** be intelligent?

What is Artificial Intelligence?

Definition?

"A machine is considered intelligent if it can perform **tasks** which are considered intelligent when carried out by a human being."



But, one necessary condition for considering a task intelligent is that it is genuinely human – and cannot be carried out by an animal and even less a machine

Goals of AI

1. Theoretical:

- Build models of how humans think and act
- Understand ourselves
- The Turing test

2. Practical:

- Replace human activities that demand intelligence
- Assist us by doing boring jobs



Main approaches to AI

- Expert systems
- Dominated AI books until the end of the XX century

Rule-based

- Logic, deduction
- Explicit coding of knowledge as formulas or rules

Machine learning

- Induction rather than deduction
- Adapt to the environment •

Now dominant

- Main focus in this course

[Source](#)

Traditional areas of AI

- Problem solving
- Search
- Game playing
- Knowledge and Reasoning

- Logic
- Theorem proving
- Knowledge representation
- Planning
- Learning
- Natural language understanding •
- Natural language generation •
- Perception
- Motion and manipulation



AI, Machine learning, Deep learning

AI ML Deep learning

Human vs Machine – Who is smartest?

30

Some tasks machines are better at •



Math/logic

31

Some tasks machines are better at

- Math/logic
- **Board games (chess, go, etc)**



Some tasks machines are better at

- Maths/logic
- Board games (chess, go, etc).
- **Some complex computer games**
 - Starcraft II and Dota 2 have been the focus of a lot of research in Reinforcement Learning
 - Minecraft is also a popular



testbed due to its open ended nature.





Some tasks humans are better at

- **Communicating with other humans**



Large language models: Towards

human level language understanding?



- TODO: First impressive example

<https://bigthink.com/the-future/chatgpt-microbiology-quiz-aced/>

But there are still important limitations



Chat with Bing Chat January 2023



40

Some tasks humans are better at

- Communicating with other humans



42

- **Moving around**

Some tasks humans are better at

- Communicating with other humans
- Moving around
- **Understanding from vision**

- **Machines are getting close to human performance**



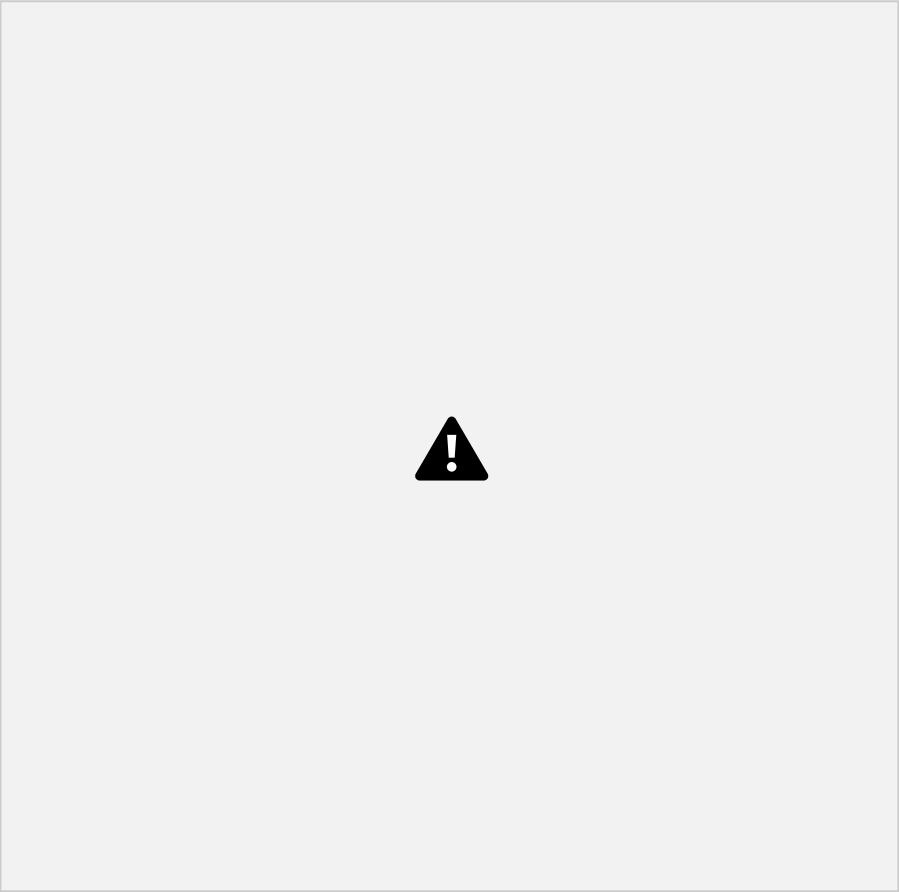


Some tasks humans are better at

- Communicating with other humans
- Moving around
- Visual tasks
- **Creative tasks**
 - **Machines are improving**

A painting of a fox sitting in a field at sunrise in the style of claude monet

Dall-E 1 (2021) Dall-E 2 (2022) Dall-E 3 (2023)





AI for music is also coming



51

Moravec's Paradox: What is easy for humans
is hard for machines

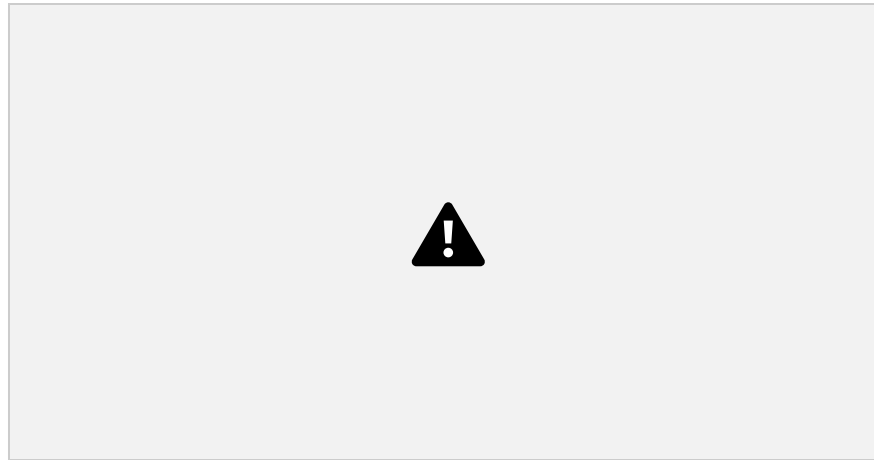
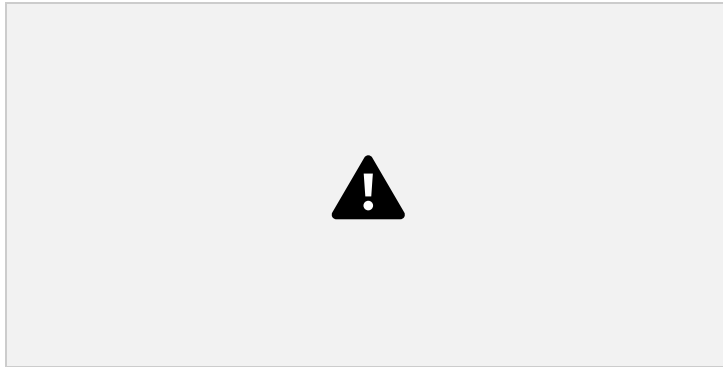


- Robotics researcher Hans Moravec (1988):

«it is comparatively **easy** to make computers exhibit adult level performance on **intelligence tests or playing checkers**, and **difficult** or impossible to give them the skills of a one-year-old when it comes to **perception and mobility**»

Moravec's Paradox

Humans^{Machines}





Next week: Search/Optimization

- Optimization: A principle behind most of the other topics in this course
- We'll talk about what optimization problems are and

begin to think about ways to
solve them

- You will also test this yourselves
in Mandatory Exercise 1

